

Impact of artificial neural networks training algorithms on accurate prediction of property values

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Abstract

This study extends the use of artificial neural networks (ANNs) training algorithms including the Powell-Beale conjugate gradient (PBCG), scaled conjugate gradient (SCG) in mass appraisal as a first attempt. The goal is to verify the comparative performance of ANNs with the linear, semi-log and log-log models. The methodologies are applied to a data of 3232 sales transactions of single-family dwellings sold during the period, January 2012 to May 2014 in Cape Town, South Africa. The results reveal that the semi-log model and the Levenberg-Marquardt trained artificial neural networks (LMANNs) performed best in their respective categories. Furthermore, the two best performed models were subjected to three different tests namely, prediction accuracy within the 10 and 20% model performance & reliability ranking order and explicit explainability ranking order. The LMANNs is found to outperform the semi-log model in the first two tests, but under the explicit explainability ranking order test, consisting of simplicity, consistency, transparency, locational and applicability within the mass appraisal environment, the LMANNs failed. The results demonstrate the semi-log model as the most preferred technique due to its simplicity, consistency, transparency, locational advantage and ease of application within the mass appraisal environment. The findings show that having better predictive power is imperative, but most importantly is whether the model can practically and effectively be used in mass appraisal of properties. The black box nature of ANNs inhibits the production of sufficiently transparent estimates required as a defence before a tribunal or in a formal court.

Key words: ANN training algorithms, mass appraisal, predictive accuracy, model transparency

1.0 Introduction

There have been concerted efforts worldwide to deal with problems of inconsistencies that sometimes occur in mass appraisal assessment. These challenges could sometimes be the result of appraisers' intuitiveness and the approach or methods employed. The intuition of an appraiser is predicated on the foreknowledge of market reactions relating to different classes of properties over time. The intuition is what influences the appraiser(s) choice of method for estimation of property values. In practice income, market and cost approaches are the most widely used appraisal methods for determination of market values of residential properties. These methods are used for single and mass appraisal until obvious limitations were observed in them. The limitations relate to subjectivity in dealing with a number of properties, delay in reporting value estimates to clients and insufficient number of comparable properties. It has conservatively been established that in mass appraisal predictions, high technology is required to comparatively handle a number of properties (McGreal, Adair, McBurney & Patterson, 1998: 57). This is to complement the human appraiser's efforts in appraising values, reduce workload and increase precision of estimates.

Several high technology methods including support vector machines, hedonic regression models, expert systems, fuzzy logic and artificial neural networks are used in mass appraisal. The hedonic regression models have been the most extensively used techniques in modelling property prices among practitioners and academics (Zurada, Levitan & Guan, 2011: 350). The hedonic regression models are used to advise

mortgage lenders, local tax authorities, dissolved companies etc. on market values of properties. However, despite the widespread use, the methods are faced with a number of shortcomings including inability to handle specification error exacerbated by nonlinearity, multicollinearity and functional form (Do & Grudnitski, 1992: 38; Worzala, Lenk & Silva, 1995: 185). The shortcomings led to the emergence of a number of propositions towards the use of non- or semi-parametric regression techniques in mass appraisal. The artificial neural networks (ANNs) are among the techniques designed to remediate the obvious limitations of the hedonic regression models. Pioneering works in the field included Borst (1991); Do & Grudnitski (1992); Tay & Ho (1992); Borst (1995); Worzala, Lenk & Silva (1995) and McCluskey (1996). The model is designed to handle the complex nonlinear relationship that exists in a data without the many parametric restrictions that are found in the statistical techniques.

Numerous elements and processes are required for the smooth operation of the ANNs. Training the ANNs is one of such processes that are fundamental towards the achievement of desired results. The purpose of the training phase is to reduce a cost function usually defined as sum squared error (SSE), mean squared error (MSE) or root mean squared error (RMSE) between actual and the predicted property sale values by adjusting weights and biases. There exists a number of training algorithms including back propagation (BP), Levenberg-Marquardt (LM), Powell-Beale conjugate gradient (PBCG) and scaled conjugate gradient (SCG). The most frequently used training algorithm in mass appraisal of properties is the BP, first developed by Werbos (1974) and popularised for multilayer perceptron by Rumelhart, Hinton & Williams (1986). Most of the known studies involving the ANNs utilise the BP training algorithm (Borst, 1991; Do & Grudnitski, 1992; Tay & Ho, 1992; Borst, 1995; Worzala, Lenk & Silva, 1995; McCluskey, 1996; Lenk, Worzala & Silva, 1997; McGreal, Adair, McBurney & Patterson, 1998; Nguyen & Cripps, 2001; Limsombunchai, Gan & Lee, 2004; Peterson & Flanagan, 2009; McCluskey, Davis, Haran, McCord & McIlhatton, 2012: 275; and McCluskey, McCord, Davis, Haran, & McIlhatton, 2013). Additionally, there has been no definite conclusion on the superiority of the ANNs over the hedonic regression models in most of these studies. Furthermore, the BP algorithm has an unimpressive behaviour of getting entrapped in local minima, longer iterations and parameter settings that are user dependent. Therefore, in this study two ANN training algorithms are proposed in mass appraisal. The goal is to compare their performance relative to previously used training algorithm and the hedonic regression models as a first attempt.

Accordingly, the methodologies are applied to a data of 3232 sales transaction of single-family dwellings in Cape Town, South Africa. The study concludes that training the ANNs with other algorithms namely LM, PBCG and SCG considerably improves the performance of ANNs than the BP but the LM outperforms all algorithms. The study also shows the relevancy of the hedonic regression models in the mass appraisal environment despite its underperformance to the ANNs. The study is organised into six sections including section one providing the background leading to the study. Section two provides the literature review, section three gives an overview of the methodology and training algorithms, section four outlines the description of the data and procedures used in the research, section five establishes the baseline regression models for the Cape Town property market, applies the training algorithms and carry out a comparative analysis of all models, while section six contains some concluding remarks.

2.0 Literature review

2.1 Pricing of properties with traditional hedonic regression

According to Goodman & Thibodeau (1998: 122) the hedonic regression technique has given crucial procedures for analysing commodities that had previously seemed extraordinarily complex. The real property fell under the category of previously hard to analyse commodities but the work of Rosen (1974) provided a framework for real estate pricing. The central idea behind the hedonic regression is that different unit (attributes) of a property are aggregated to develop a price for the property. The use of this technique can be traced to Court (1939) who used it in the automobile industry and also Lancaster (1966) who relates it to the bundle of characteristics which provide utility to consumers of goods. The study of Malpezzi,

Ozanne & Thibodeau (1980) relates real property to consumer goods like bundle of groceries having differing sizes and items. Relating this to the different component features of property, Sirmans, Macpherson & Zietz (2005: 4) reported that property is a bundle of structural, environmental and spatial attributes. These attributes differ one from the other property, therefore the hedonic multiple regression would price the features on a collective sample of many dwellings.

The study of Des Rosier & Thériault (2008: 113) listed a number of areas that the model had previously been used and found the model to particularly be important to the property market because of high level of competition that is open to buyers and sellers. In practice, the model has been used for pricing of real property for more than four decades. Accordingly, Goodman (1978) used the hedonic regression model on a data base of 1835 single family dwellings in the New Haven, standard metropolitan statistical area (SMSA) to form submarkets and indices that determine relative prices of housing services. However, despite the robustness of this model the choice of functional form and ability to capture spatial features are among its limitations. On functional form specification, the economic theory fails to specify the functional relationship that a property price and its attributes should take. This according to Crooper, Deck & McConnel (1988: 668) led a number of scholars to rely on goodness of fit criterion as suggested in Rosen (1974); Goodman (1978) to select appropriate functional form. However, their study noted that when all attributes are observed, linear and quadratic Box-Cox transformed variables gives accurate estimates of marginal attribute prices but this change when certain variables are not observed or replaced by proxies a linear function outperformed the quadratic Box-Cox function. Again, the study of Goodman (1978: 483) found the frequently used linear form to be overly restrictive and favour the Box-Cox transformation. Several criticisms trailed the use of Box-Cos function leading to a situation where some authors directly formulates a model structure without recourse to the hedonic function (Borst, 2007: 35).

In general, apart from the basic linear multivariate specification, authors commonly used the semi-log and log-log specifications. In the semi-log formulations, the left hand side of the equation namely the dependent variable (property price) is regressed against the linear arrays of structural and locational characteristics while in the log-log formulation, both the dependent variable (property price) and independent variables (structural features) are transformed. Therefore, all assessment must at least take a particular form which researchers in practice usually specify in their model calibration. Accordingly Kang & Reichert (1987: 1) noted the significance of regression coefficients and that the prediction accuracy depend on the choice of estimating technique and the functional form of the regression equation. The study of Schulz, Wersing & Werwatz (2014) used the price and log-prices as dependent variable while other variables remain in their linear format on 18,444 single-family transactions to estimate property prices and found the semi-log hedonic regression to be a better choice. Recently the study of McCluskey *et al.* (2012) undertook a study on a 2694 property data in the Lisburn district of Northern Ireland, UK using linear, semi-log and log-log hedonic regression models and found the semi-log model to outperform other models. The three studies show that log transformation could improve the predictability of hedonic regression. However a dissimilar result was achieved with semi-log model, in a study by McCluskey (2016) on a dataset of 46,689 (before the removal of outliers) in Kazakhstan using three scenarios, that included analysis after the first data cleaning, second data cleaning and semi-logarithm, the best result (67%) was obtained after the second data cleaning with the linear additive regression. The result of the reviewed studies shows that sometime log transformation do not assured of optimal results. Therefore it is good to first undertake a test that supports the need for log transformation of variables before assessment. There are studies that find an alternative way of avoiding the parametric restrictions of the hedonic regression leading to the introduction of ANNs in modelling property prices.

2.2 The application of ANNs in mass appraisal of properties

The ANNs has been applied for mass appraisal since the 1990s in so many research papers with mixed results (McCluskey, *et al.* 2012: 275). Several papers have been written on the potential of the model within the real estate sector. While some studies found the model as a useful tool for effective prediction of

property prices others took a contrary position. For instance, earlier studies of Borst (1991), Evans, James & Collins (1992), Tay & Ho (1992), Do & Grudnitski (1992) found it useful in mass appraisal. Since the introduction of the ANNs is premise on overcoming the functional and nonlinear limitations of the hedonic regression, most studies compare the prediction accuracy of the ANNs relative to the hedonic regression in different countries. Accordingly, studies such as Tay & Ho (1992) undertaken in the Singapore; Do & Grudnitski (1992) in the USA; Evans *et al.* (1992) carried out in the UK; Borst (1995) undertaken in the USA all establish the fact that ANNs performed better than the hedonic regression models. Similarly, studies such as Nguyen & Cripps (2001) carried out in the USA; Limsombunchai, Gan & Lee (2004) undertaken in the New Zealand; Peterson & Flanagan (2009) also in the USA all support earlier studies that ANNs is better than the hedonic regression models. However, a contrary position was taken in studies of Worzala, Lenk & Silva (1995); Lenk, Worzala & Silva (1997); and McGreal, Adair, McBurney & Patterson (1998). These studies also gave a caveat to the assessment community against the use of ANNs in mass appraisal because of the unreliable results establish in their studies.

Also the study of Zurada *et al.* (2011) that utilises hedonic regression, ANNs, additive regression (AR), M5P trees, SVMs, radial basis function neural networks (RBFNN) and memory-based reasoning (MBR) in Louisville, Kentucky, USA found that AR, M5P trees and SVMs performed better than the ANNs, RBFNN and MBR models. Lin & Mohan (2011) in Amherst, New York, utilised three methods namely hedonic regression, ANNs and ANR in mass appraisal. The study found the ANNs to be better than the hedonic regression and the ANR in both training and validation data. McCluskey *et al.* (2012) in Lisburn, Northern Ireland undertook a study using the linear, semi-log and log-log models on the one hand and the ANNs on the other and found the three regression models to predict property prices better than the ANNs. Again, McCluskey *et al.* (2013) carry out a comparative study on the predictive accuracies of hedonic regression model, simultaneous autoregressive (SAR), geographically weighted regression (GWR) and ANNs and found the GWR to perform better than other models.

The studies of Peterson & Flanagan (2009); McCluskey *et al.* (2012) are of significant interest to this current study because of the RESET test carried out to support the case for the use of the nonlinear models. Peterson & Flanagan (2009) observe that the hedonic regression models are opened to pricing errors owing to the way means are extrapolated from large samples and as such will also be exposed to significant sampling errors. The study also noted that specification error is also unavoidable in *ad hoc* specification and to the extent that value does not map linearly on to property characteristics, so too are errors dues to neglected nonlinearities. Again to the point that nonlinear models integrate linear forms, then nonlinear models would be the desired choice. However, the exact nonlinear form is neither apparent nor are their practical steps one would take to find the correct form. The ANNs do, however, provide a practical alternative to conventional least squares form (including nonlinear least squares) that is easily implementable and which efficiently models nonlinearity in the underlying relationships (including the parameters) Peterson & Flanagan (2009).

The above studies provided inconsistent results that reveal different performance of models. In McCluskey & Anand (1999: 221) a summary of different methods was provided revealing their strengths and weaknesses concluding that almost all methods have limitations in terms of prediction accuracy, transparency, and ease of application. In Kauko & d'Amato (2008: 17), these limitations were observed but suggested that certain pertinent issues should be reflected upon. These are: is the challenge on the accuracy of techniques? Is it on the feasibility of using the model in mass appraisal environment or on some non-technical portion that require a more detailed analysis, such as making adjustment to the model structure? It is expected that when these issues are reflected upon, some improvements or modifications should be implemented so that optimality could be realised in a model(s). Such improvement may be effected on the model structure or data quality.

Accordingly, McCluskey *et al.* (2012) utilised the different regression models having observed its limitations in handling nonlinearity in property data relative to the ANNs. The current study extends their

work by improving on the structure of the ANNs in the field of mass appraisal. The ANNs have a structure that permits the adjustment of weights on the basis of patterns in the property dataset. Should discrepancies occur, the weights are altered changing the original state of the network, so the system appears to learn (McCluskey, 2012: 275). This alteration is performed with the aid of the back propagation (BP) learning algorithm. BP uses the gradient descent search method to modify connection weights in order to minimise error between actual and desired output. The algorithm is simple in execution and suitable for solving different problems. However, foremost among its challenges are inability to continuously achieve global optimum, longer iteration and possibility of getting entrapped into local optima.

3.0 Methodology and algorithms

3.1 Hedonic Regression Models

Janssen & Söderberg (1999: 361) reports that the hedonic regression gives the framework for the assessment of “differentiated goods like housing units whose individual features do not have observable market prices”. The model takes each attribute of a property, measure their contribution and assign a weight for it. These attributes when aggregated together will produce the price of a property. The ability of the hedonic regression to capture complex interaction that exists within the property variables and provide estimates in an explicit and transparent manner makes it attractive, acceptable and widely used for property tax assessment (McCluskey *et al.*, 2012: 277; McCluskey *et al.*, 2013: 246). However, McCluskey *et al.* (2012: 278) noted that three important elements must be addressed for effectiveness of this model including careful selection of dependent and independent variables; choice of functional form and lastly the statistical relevance and contribution of the independent variables to the model. In this study the dependent variable is the property assessed values while the independent variables were selected based on the *a priori* expert knowledge that they are likely to have positive influence on property prices. The functional form to be used in hedonic modelling is not specified in the economic theory; therefore in this study the linear, semi-log and log-log models are used. The study of Palmquist (1979: 442) pointed out that selection must be approached from an empirical viewpoint. The general formulation of a linear model is given in equation 3.1

$$Y = b_0 + b_1x_1 + b_2x_2 + \dots + b_sx_s + \varepsilon \quad (3.1)$$

or

$$Y = b_0 + \sum_{i=1}^s b_ix_i + \varepsilon$$

where Y is the assessed values of properties; $x_1, x_2, \dots, x_s$ are the independent variables; $b_1, b_2, \dots, b_s$ are coefficients or prices per unit assigned by the model to the independent variables; b_0 is the regression constant and ε is the error term. To model a nonlinear interaction between variables the semi-log and log-log models are given in equations (3.2) and (3.3)

$$\text{Log}Y = b_0 + b_1x_1 + b_2x_2 + \dots + b_sx_s + \varepsilon \quad (3.2)$$

and

$$\text{Log}Y = b_0 + b_1\text{Log}x_1 + b_2\text{Log}x_2 + \dots + b_s\text{Log}x_s + \varepsilon \quad (3.3)$$

The importance of log transformation is noted in Schulz, Wersing & Werwatz (2014: 133). The study observes that using the log transformation could partially remediate the inherent heteroscedasticity of property prices. It also makes the effective relationship among variables nonlinear while still preserving the linear model. Furthermore, Gloudemans (2002: 26) report that of the three functional forms, the linear additive model is widely used for single-family dwellings, easy to calibrate and allows the contribution of

each variable to be added to the model. However, foremost among its limitations are inability to deal with the problems of spatial dependence and spatial heterogeneity which are commonly found in property markets. This study concentrates on dealing with the functionality and nonlinear limitations of hedonic regression models through the use of ANNs. Since the ANN models have the capability of recognising patterns, the influence of spatial effects on property data could easily be suppressed.

3.2 Artificial Neural Network Models

According to Engelbrecht (2007: 5) the human brain is a complex, nonlinear and parallel computer with the ability to handle tasks at a greater speed such as pattern recognition, perception and motor control than any computer, irrespective of the shortness of time or moments at which event occur for neural systems. The ability of the human brain to learn, memorise and generalise well created research interest to artificially mimic the biological neural systems (BNSs) commonly known as the ANNs. The BNSs are nerve cells also referred to as neurons containing the cell body, dendrite and an axon massively interconnected. The connection between the axon of the neuron and the dendrite of another neuron is commonly referred to as a synapse (Figure 3.1). The signals are conveyed from the dendrites through the cell body to the axon from where the signals are transmitted to all connected dendrites.

The artificial neuron is modelled after the biological neuron of the human brain. The artificial neurons (ANs) are units in the ANNs that receive one or more inputs as numerical values associated with their respective weights. A bias or threshold level is added as an additional input value to the summation function. The summed value is passed to the next phase to execute the activation function which produces the output from the neuron as shown in Figure 3.2. The detail processes of the ANNs are given in the next paragraph.

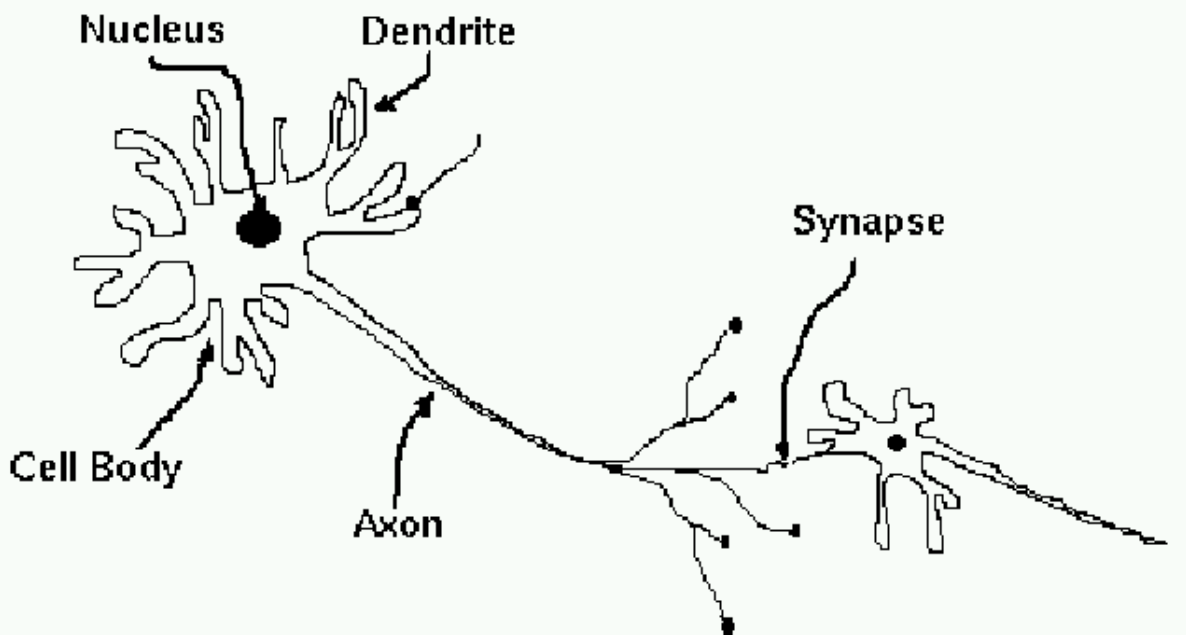


Figure 3.1 A biological neurons (adopted from Engelbrecht (2007: 6)).

The ANNs has multilayer perceptron as the most popular network architecture, consisting of an input layer, an output layer and at least a hidden layer for processing the nonlinear elements. The least (one) number of hidden layers is a reminiscence of the suggestion given in Masters (1993) that a hidden layer should be the initial choice for any practical ANNs design. In this regard, the study of Lin & Mohan (2011: 234) revealed that a single hidden layer is adequate for ANNs to approximate any complex nonlinear function and achieve accuracy (Hornik, 1991). In this study a hidden layer was used to construct the architecture of the ANNs.

The values of the input variables are sent into the network via the input layer to the hidden layer. The number of input variables in the network is determined by the configuration of the input data. The output layer is fully connected to the neurons of the input layer and this line of connection runs through the entire units of the network.

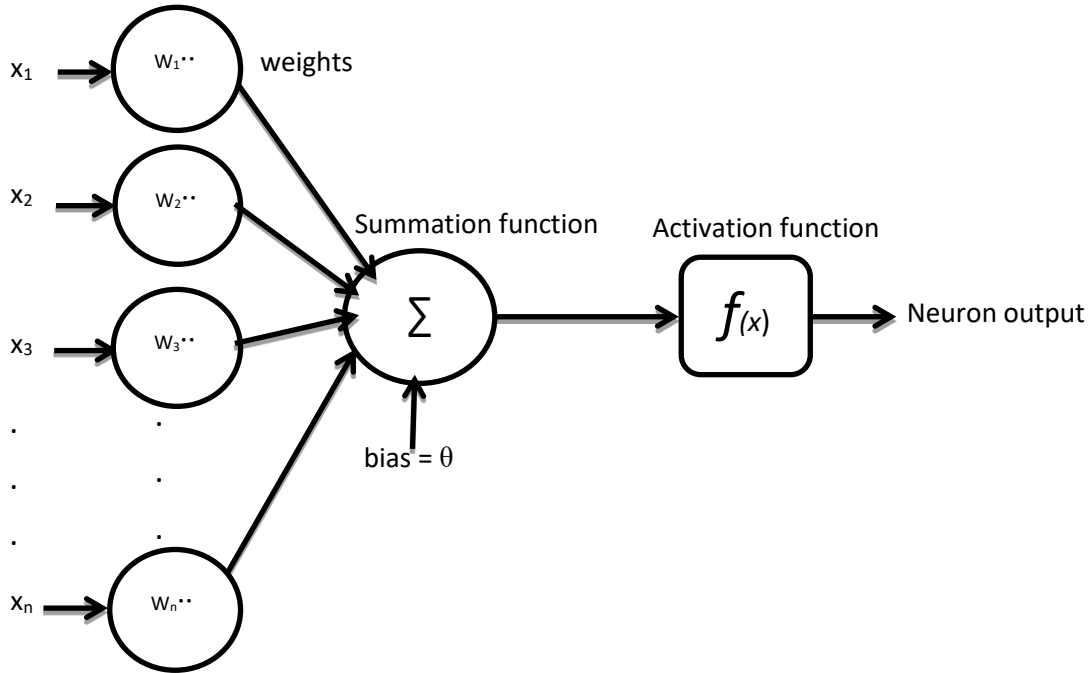


Figure 3.2 The structure of the artificial neuron.

The number of neurons in the hidden layers is a matter of user discretion through a trial and error process. Kwok & Yeung (1997: 631) proposed that the method of finding optimal number of hidden neurons is to begin with a lesser number of neurons, the number should be increased until a desired result is found. The same procedure of optimal selection of the number of hidden neurons as used by Lin & Mohan (2011: 234) and McCluskey *et al.* (2013: 250) was employed in this study. Another important element of the neural networks is the transfer function that determines the relationship between inputs and output (target) of the neuron and its network. The preferred transfer function in this study was the tan-sigmoid used in the neurons of the hidden layer and the linear transfer function was used in the neurons of the output layers. The network compares the output with the actual value to ascertain its accuracy by the total mean squared error. In building a model, the usual practice is to split the data into two or three such as training, testing and validation.

3.2.1 Artificial Neural Networks Training Algorithms

The ANNs utilise a number of training algorithms including conjugate gradient (CG), LM, and BP being the most widely used, amongst many others. The discussion in the next few paragraphs will dwell on them.

3.2.1.1 Back Propagation Algorithm (BP)

The BP algorithm is not the first training rule for ANNs (Engelbrecht, 2007: 24). However, in most real estate mass appraisal assessment the BP is the most commonly used algorithm in training the ANNs. The sum squared error given in equation (3.4) is required to evaluate the training process.

$$E(y, w) = \frac{1}{2} \sum_m^M \sum_l^L (d_{lm} - o_{lm})^2 \quad (3.4)$$

where y and w are the network input and weight vectors, m is the index of patterns, from 1 to M , in which M denotes the number of training patterns; l is the index of outputs, from 1 to L , in which L is the total number of outputs; k and p are indices of weights, from 1 to N , where N is the total number of network weights and d_{lm} and o_{lm} are desired and actual values of the m^{th} output and the l^{th} patterns. Yu & Wilamowski (2011) noted that the gradient g is defined as the first-order derivative of the total error function (equation 3.4) as

$$g = \frac{\partial E(y, w)}{\partial w} = \left[\frac{\partial E}{\partial w_1} \quad \frac{\partial E}{\partial w_2} \quad \dots \quad \frac{\partial E}{\partial w_N} \right]^T \quad (3.5)$$

With the definition of g in equation (3.5), the update rule of the gradient/steepest descent algorithm is written as:

$$w_{r+1} = w_r - \alpha g_r \quad (3.6)$$

where r is the index of iterations and α is the learning constant (step size taken in the negative direction of the gradient). The training process of the gradient descent algorithm is asymptotic (approaching a curve arbitrarily closely) convergence. The convergent behaviour of BP is dependent on the choice of the initial values of the network connection weights which is related to the network parameters including the learning rate and momentum. This procedure could sometimes be problematic because of the tendency to over train the network and slow convergence. Training speed could significantly be increased by the use of second-order algorithms including Newton and LM, scaled conjugate gradient and Powell-Beale amongst others. Additionally, Openshaw (1998: 1864) suggests that another way of easing this problem might be to use meta-heuristics (GA, PSO etc.) to design and train ANNs. The use of these training algorithms is amongst others the assignment of this study.

3.2.1.2 Levenberg-Marquardt Algorithm (LM)

LM was principally designed by Levenberg (1944) and Marquardt (1963) to provide a numerical solution to nonlinear least squares problems. This algorithm has the properties of gradient descent stability (slight variation in training error even if there is a change in training data point) and Gauss-Newton (GN) speed which gives it advantage over all others. These attributes are what carved a unique niche to LM as a versatile ANNs training algorithm because in many respect convergence speed is assured despite complexities of the error surface. According to Yu & Wilamowski (2011a) the central idea of LM is that its training process is combined, switching between two algorithms. Therefore, if the combination coefficient μ is small, GN algorithm is utilised but when combination coefficient is large, the gradient/steepest descent algorithm is employed. Consequently, an adjustment is made to μ parameters so that the network converges to an optimum desired position. The GN algorithm simplifies the calculation of the second-order derivative while the LM is considered a trust-region that modifies the GN method (Battiti 1992: 159).

The sum squared error (E) in equation (3.4) is used to evaluate the training process in LM. The update rule of LM is given as (Yu & Wilamowski, 2011: 465b).

$$\Delta w_r = (H_r + \mu I)^{-1} g_r \quad (3.6)$$

where I is the identity matrix, g is the gradient vector, the Hessian matrix is given as H and μ is the combination coefficient. The Hessian matrix (H) and gradient vector (g) are defined in equations (3.7) and (3.8) as

$$H = \begin{bmatrix} \frac{\partial^2 E}{\partial w_1^2} & \frac{\partial^2 E}{\partial w_1 \partial w_2} & \dots & \frac{\partial^2 E}{\partial w_1 \partial w_N} \\ \frac{\partial^2 E}{\partial w_2 \partial w_1} & \frac{\partial^2 E}{\partial w_2^2} & \dots & \frac{\partial^2 E}{\partial w_2 \partial w_N} \\ \dots & \dots & \dots & \dots \\ \frac{\partial^2 E}{\partial w_N \partial w_1} & \frac{\partial^2 E}{\partial w_N \partial w_2} & \dots & \frac{\partial^2 E}{\partial w_N^2} \end{bmatrix} \quad (3.7)$$

$$g = \begin{bmatrix} \frac{\partial E}{\partial w_1} \\ \frac{\partial E}{\partial w_2} \\ \dots \\ \frac{\partial E}{\partial w_N} \end{bmatrix} \quad (3.8)$$

According to Yu & Wilamowski (2011: 466b) the second order derivative of E in equation (3.7), has to be computed in order to perform the update rule in equation (3.6). The process according to Battiti (1992: 148), Yu & Wilamowski (2011a), Yu & Wilamowski (2011: 466b) is complicated. Furthermore, in the Hagan & Menhaj (1994: 990) and Yu & Wilamowski (2011: 466b) implementation of the LM algorithm, Jacobian matrix (J) was introduced so that the process of computation will be simplified by avoiding the second-order derivatives as

$$J = \begin{bmatrix} \frac{\partial e_{11}}{\partial w_1} & \frac{\partial e_{11}}{\partial w_2} & \dots & \frac{\partial e_{11}}{\partial w_N} \\ \frac{\partial e_{12}}{\partial w_1} & \frac{\partial e_{12}}{\partial w_2} & \dots & \frac{\partial e_{12}}{\partial w_N} \\ \vdots & \vdots & \dots & \vdots \\ \frac{\partial e_{1L}}{\partial w_1} & \frac{\partial e_{1L}}{\partial w_2} & \dots & \frac{\partial e_{1L}}{\partial w_N} \\ \vdots & \vdots & \dots & \vdots \\ \frac{\partial e_{M1}}{\partial w_1} & \frac{\partial e_{M1}}{\partial w_2} & \dots & \frac{\partial e_{M1}}{\partial w_N} \\ \frac{\partial e_{M2}}{\partial w_1} & \frac{\partial e_{M2}}{\partial w_2} & \dots & \frac{\partial e_{M2}}{\partial w_N} \\ \vdots & \vdots & \dots & \vdots \\ \frac{\partial e_{LM}}{\partial w_1} & \frac{\partial e_{LM}}{\partial w_2} & \dots & \frac{\partial e_{LM}}{\partial w_N} \end{bmatrix} \quad (3.9)$$

By combining equations (3.4) and (3.8), the gradient vector elements can be calculated as

$$\frac{\partial E}{\partial w_k} = \sum_{m=1}^M \sum_{l=1}^L \left(\frac{\partial d_{lm} - o_{lm}}{\partial w_k} d_{lm} - o_{lm} \right) \quad (3.10)$$

So that the relationship between Jacobian matrix (J) and gradient vector (g) can be presented by

$$g = J^T e \quad (3.11)$$

By combining equations (3.4) and (3.7), the Hessian matrix (H) can be computed as

$$\begin{aligned}
\frac{\partial^2 E}{\partial \mathbf{w}_k \partial \mathbf{w}_p} &= \sum_{m=1}^M \sum_{l=1}^L \left(\frac{\partial d_{lm} - o_{lm}}{\partial \mathbf{w}_k} \frac{\partial d_{lm} - o_{lm}}{\partial \mathbf{w}_p} + \frac{\partial d_{lm} - o_{lm}}{\partial \mathbf{w}_k} d_{lm} - o_{lm} \right) \\
&= \sum_{m=1}^M \sum_{l=1}^L \frac{\partial d_{lm} - o_{lm}}{\partial \mathbf{w}_k} \frac{\partial d_{lm} - o_{lm}}{\partial \mathbf{w}_k}
\end{aligned} \tag{3.12}$$

So that the relationship between Jacobian matrix (J) and Hessian matrix (H) can be described by equation (3.13) as

$$H = J^T J = Q \tag{3.13}$$

where matrix Q is the approximated Hessian Matrix (H) known as quasi Hessian Matrix. Therefore the LM update rule is achieved by integrating equations (3.6) and (3.10) with equation (3.13).

$$\Delta \mathbf{w}_r = (J_r^T J_r + \mu I)^{-1} J_r^T e_r \tag{3.14}$$

The error vector is (e). To implement the LM algorithm, equation (3.14) is used. The procedure is to first calculate and store Jacobian matrix, then carryout matrix multiplications of equations (3.11) and (3.13) for more weight update. In the Jacobian matrix (J) defined in (3.23), there are $M \times L \times N$ elements that needed to be stored. Wilamowski, Kaynak, Iplikci & Efe (2001: 1779) and Yu & Wilamowski (2011: 467b) report that the LM algorithm is effective for solving problems with small and medium sized training patterns but ineffective in handling large sized training patterns because of its memory limitation. To support this view, Wilamowski & Yu (2010: 931) observe its limitation in solving parity-16 problem of 65,536 patterns and suggests an improvement of not storing Jacobian matrix but replacing Jacobian matrix multiplication with vector operations so that problems with unlimited number of training patterns are solved. The size of training patterns in this study is considered adequate (small) for the LM algorithm, however, for a large size pattern the approach suggested and used in Wilamowski & Yu (2010) should be the preferred.

LM has been used in many fields for ANNs training with optimal results. The only mass appraisal/valuation study found that involved LM is the study of El Hamzaoui & Perez (2011) undertaken in Casablanca, Morocco. The study is of exploratory nature and found the algorithm as a useful model in ANNs training. This research is considered to be more comprehensive in the analysis of residential properties to ascertain its capability in ANNs training.

3.2.1.3 Conjugate Gradient Algorithm (CG)

Unlike BP algorithm that adjust weights in the direction of the negative gradient of the error surface, conjugate gradient algorithm performs searches in the conjugate paths which usually gives convergence that is faster than searches made along gradient descent directions. The algorithm trades off the simplicity of gradient/steepest descent and fast quadratic properties of Newton's method (Engelbrecht, 2007: 45). Many variants to conjugate gradient algorithm used in training feed-forward ANNs including Fletcher-Reeves CG, Polak-Ribière CG, Powell-Beale Restarts CG and Scaled Conjugate Gradient exists. In this study, Powell-Beale Restarts and scaled conjugate gradient are the primary focus. This is because the Powell restart method gives an opportunity of restarting if there is little orthogonality (shift or change) left between the current and previous gradient (Sharma, Sharma & Kasana, 2007: 1116). Again, to save time which is akin to CG during line search at each iteration step, the scaled conjugate gradient utilises a step size scaling approach for line search per iteration.

3.2.1.3 Scaled Conjugate Gradient Algorithm (SCG)

Scaled gradient algorithm is another approach used in estimating the step size (Møller, 1993: 529). This algorithm is unique in training feed forward ANNs because it combines LM with conjugate gradient algorithms. Accordingly, the process involves introduction of a scalar parameter denoted as β_r in CG, where r represents $0, 1, 2, 3, \dots, N$, the step size α_r is positive and the path p_r are generated by the equation

$$p_{r+1} = \theta_{r+1} g_{r+1} + \beta_r p_r \quad (3.15)$$

This is the search path, also known as scaled conjugate gradient algorithm. In equation (3.15), parameters that are to be determined are θ_{r+1} and β_r . Thus if $\theta_{r+1} = 1$, the result is a classical conjugate gradient algorithm based on the value of scalar parameters β_r (Andrei, 2007: 403). Conversely, if scalar parameter is 0 ($\beta_r = 0$), then another class of algorithm that supports the choice of parameter θ_{r+1} evolved. Looking at $\beta_r = 0$, it is possible for θ_{r+1} to assume a positive scalar or negative definite matrix. These possibilities are either a change in the paths of gradient descent or Newton algorithms. If $\theta_{r+1} = 1$, a gradient descent algorithm ensued. But if $\theta_{r+1} = \nabla^2 f(x_r)^{-1}$, or estimation of it (Andrei, 2007: 403), then it shifts to the path of Newton or Quasi-Newton algorithms. Again if $\theta_{r+1} \neq 1$ is chosen in a quasi-Newton way and $\beta_r \neq 0$, equation (3.15) characterises a blend of quasi-Newton and conjugate gradient techniques. According to Andrei (2011: 325) if θ_{r+1} is a matrix that has information about inverse Hessian of function f , then it is better to use $p_{r+1} = -\theta_{r+1} g_{r+1}$ because adding the term $\beta_r p_r$ might affect the path p_{r+1} from being a descent path except if the line search is accurately sufficient.

Scaled conjugate descent algorithm has been applied in ANNs training for parity problems using 20 different initial weight vectors with a momentum set at 0.9 for all simulations. The result indicates that SCG converges faster than standard BP, conjugate gradient with line search and Brayden-Fletcher-Goldfarb-Shanno memory-less quasi-Newton algorithms (Møller, 1993: 532). Orozco & García (2003: 354) used SCG to avoid time consuming line search per iteration of other second order CG algorithms in classifying two types of infants cry. Also Ambarish & Saroj (2016: 500) applied the SCG algorithm in the prediction of soil moisture content for the control of farm irrigation in eastern India. Their study found that SCG performed better than Broyden Fletcher Goldfarb Shanno (BFGS) in training feed-forward ANNs. With this in mind, SCG is proposed for ANNs training in mass appraisal of properties.

3.2.1.4 Powell-Beale Conjugate Gradient Algorithm (PBCG)

In conjugate gradient algorithm the search path is reset to the negative position of the gradient periodically. When number of weights and biases in the network equates the number of epochs, the standard reset point is said to occur, however, other reset approaches such as Powell-Beale are used to influence the efficiency of network training. These approaches are proposed by Powell (1977) based on the earlier work of Beale (1972). Powell's proposition is that a restart should ensue if the orthogonality between the new and the former gradient is left to a very minimal amount. This is defined by the following condition to reset the steepest descent path

$$\left| g_{r-1}^T g_r \right| \geq \alpha \|g_r\|^2 \quad (3.16)$$

where α is the restart/reset factor which has a range of 0.1, 0.2, ... 0.9 (Powell, 1977: 251), g is the gradient and subscript r is the gradient index. The major shortcoming of this algorithm is the expensive nature of computation of line search requirements for each iteration step. This algorithm has been utilised in ANNs training for identifying the presence of red palm weevil (Al-Saqer & Hassan, 2011: 362). Specifically their study shows that although Powell-Beale conjugate gradient training algorithm consumes less training time, SCG algorithm outperformed Powell-Beale CG in their assessment.

4.0 Methods and procedures

4.1 The Cape Town assessment

The city of Cape Town is 944 square miles; it is located on a peninsula beneath the magnificent Table Mountain on the southwest coast of South Africa. It is the provincial capital of the Western Cape and the legislative capital of the country. There are other municipalities that are adjacent to it including Swartland and West Coast to the north; drakenstein, Cape Winelands and Stellenbosch to the north-east; and Theewaterskloof, Overberg and Overstrand to the south-east. The city of Cape Town has centralised its property tax assessment under the city valuation office (CVO). The CVO is the body responsible for the valuation of 915,148 residential and commercial properties in Cape Town (KPMG, 2015: 12). The CVO has created 1555 neighbourhoods and 31 submarkets for the purpose of assessment and reassessment.

4.2 The data

The data used for this analysis was supplied by the city valuation office Cape Town, South Africa. The data contains some missing, extreme values and unreliable transactions that were removed because leaving them could affect effective assessment. In the original data, there are 46 property variables and 3526 but after the removal of questionable transactions, cost related and variables that simply describe the property such as single family dwellings, 3232 property transactions and 11 variables were left for analysis. The sample was considered adequate for the analysis because the methods themselves are not in any case controlled by the number of properties that can be handled (McCluskey & Anand, 1999: 231). There are texts data such as property quality, condition, building style, and property view, which were renamed and assigned numeric values for ease of assessment. The assignment of numeric values is done such that for say property quality, 6 bands exists reflecting poor, fair, average, good, very good and excellent factors which were assigned the values of 1 and 6, respectively. The same procedure was used for all other text data (condition, building style, and property view).

Furthermore, in the variable building style, a dummy variable “group housing” (style_group) was found in one transaction, this was added to style_medit/t category. Transactions above 4 storeys were included in the modelling activity but recoded as 3 storeys because they are few in number. Transactions with zero storeys were considered ground floor or bungalow. The two transactions found in the category were recoded to reflect one storey category. The recoding is consistent with the study of Guan, Zurada & Levitan (2008: 403–406) designed to ameliorate the problem of dimensionality which is capable of reducing the strength of a model. Depending on the platforms, binary, continuous and categorical variables were used in this analysis. The variables were left in their categorical and continuous states when the ANNs were used but when the hedonic regression models were applied, the binary dummy format was used. This became necessary because of the limitation of ANNs in the use of excessive dummy variables (McCluskey *et al.*, 2013: 253; Feng & Jones, 2015: 110). For instance, the variables condition and quality were coded between 1 and 5 or 6 when the ANNs is used but when the OLS, semi-log, log-log models are used this was coded 1 and 0 (with 1 if the categorical condition is met, 0 if otherwise) leaving out the most commonly occurring category to avoid the dummy variable trap that might lead to a condition of perfect multicollinearity ensuing the failure of the regression program (Greene, 2003: 118; Borst, 2007: 50). Table 4.1 contains the summary of the property data which are deemed useful for this analysis.

Table 4.1 Property variables description

Variable	Description
<i>Assessed_values</i>	Assessed values of properties in South African Rand (1Rand = USD15)
<i>Beds</i>	Total number of bedrooms
<i>Quality</i>	Quality grade factor of construction

<i>Condition</i>	Condition/state of property
<i>Storey</i>	Total number of storeys
<i>Bld_style</i>	Building design and architectural style
<i>View</i>	Quality grade factor of property
<i>RMOS</i>	Reverse month of sale
<i>Size</i>	Size of property in square metres
<i>Pool</i>	Size of swimming pool in square metres
<i>SMKT</i>	Locational variable identifying submarkets

The number of variables (11) was carefully chosen using pragmatic and realistic approaches. They are also considered likely to be value significant in line with previous multivariate analysis of this nature. Table 4.2 summarise the descriptive statistics of the property data used in this study. The statistics illustrates the variability within the data. The mean assessed value in the sample data is R4,483,474.29 while the average size of property is 177. The smallest property assessed in terms of size is 31 square metres. The average (mean) number of storey in the sample is two. The average property in the sample has four bedrooms. The total transactions were stratified into 70% training and 30% testing datasets in accordance with acceptable norm that data should be partition to allow for modelling or training and testing. The stratification was done using WEKA explorer via a resample procedure.

Table 4.2 Descriptive statistics of property variables

Variable name	Mean	SD	Minimum	Maximum
<i>Assessed_value</i>	4483474.29	3117754.039	824000	38000000
<i>Beds</i>	3.56	0.992	1.0	10
<i>Quality</i>	3.494	0.6155	1.0	6.0
<i>Condition</i>	3.506	0.6269	1.0	5.0
<i>Storey</i>	1.52	0.553	1.0	3.0
<i>Bld_style</i>	3.034	0.4300	1.0	7.0
<i>View</i>	3.581	0.9630	1.0	6.0
<i>RMOS</i>	14.885	8.1645	1.0	29.0
<i>Size</i>	177.45	78.905	31	599
<i>Pool</i>	13.97	18.362	0.0	154

The sales date (temporal or time trend) is very important in any modelling activity. This becomes necessary because of fluctuation in property prices over a time period in different geographical areas of the property market. Different techniques are employed to incorporate time trend into a model including paired sales analysis, resale analysis, incorporating time variable in regression models and using current appraised value (Gloudemans, 1990: 84). In a recent study, Borst (2008: 34–35); Borst (2009: 29–30) suggested the application of reverse month of sale (RMOS), Fourier expansion, quarterly or semi-annual binary (dummy) variable and linear spline method to capture time trend in a model. In this study the RMOS and semi-annual dummy variables were used for ANNs and hedonic regression models to account for time trend. Again while the RMOS are used for modelling with ANNs, the semi-annual dummies are used for modelling prices with the hedonic regression. The date of sale in the sample data covers a period from January 2012 to May 2014. Consequently, in using the RMOS, the most recent month in the sample is RMOS = 1; the next is RMOS = 2; and this continue to the oldest month of sale, assigned the RMOS = 29. There are five bands created to reflect the semi-annual dummies used in this study. Accordingly, the notation SA1 was used to represent the first half year that equals 1 for RMOS = 1–6 otherwise 0; SA2 was used to describe the second half year that equals 1 for RMOS = 7–12, otherwise 0. For the third half year the notation SA3 equals 1 for RMOS = 13–18, otherwise 0. This was done for all periods until the last quarter, however, to mitigate against dimensionality and dummy trap problems the most occurring category (SA4) was excluded from the analysis.

Having taken into account the temporal aspect in the data, the next is to account for the spatial nature within the data. McCluskey & Borst (2007: 313) reports that there are several techniques by which location is specified and calibrated in a model. These are using market segmentation that group properties based on the different segment, neighbourhood delineation variable as an indicator reflecting the particular neighbourhood the property belong to, accessibility measures, explicit use of location and advanced model specification methods. For further emphasis, readers are referred to McCluskey & Borst (2007). The techniques are sometimes used together with others in a single study depending on the model capability, specification and availability of location element(s) within the data. This study utilise the neighbourhood delineation variable to indicate the submarket the property is located. The neighbourhoods and submarkets were delineated by Tax Assessors for ease of identification and assessment. There are 181 neighbourhoods and 31 submarkets in the property data. The neighbourhoods were used to establish 15 submarket dummy variables leaving out the most occurring submarket (SMKT54). Other studies that used the neighbourhood delineation or submarket dummy variable to account for location are Wilhelmsson (2002: 96); Bourassa *et al.* (2007: 151) and Lin & Mohan (2011: 231–233). Wilhelmsson (2002: 96) used 13 submarket dummy variables from previously defined administrative parish to account for location in the hedonic regression model. Bourassa *et al.* (2007: 151) created 33 submarket dummies to improve their results and Lin & Mohan (2011: 231–233) created 66 dummies from neighbourhood codes to form submarkets/cluster in their analysis.

4.3 Performance measurement

The Spatial Analysis in Macroecology (SAM v4.0) and Statistical Package for the Social Science (SPSS) v21 were used for assessment of property prices with hedonic regression models. MatlabR2013b and Waikato Environment for Knowledge Analysis (WEKA, 3.6) explorer were used for property pricing with the ANNs in this study. The accuracy test statistics used to explain the performance models are root mean squared error (RMSE), squared coefficient correlation (R^2), and mean absolute error (MAE). The RMSE, MAE, and R^2 are given in the formulas:

$$RMSE = \sqrt{\sum_i (y_i - \hat{y}_i)^2 / n} \quad (4.1)$$

The RMSE defined above is the square root of the average of the squared values of the estimation error. RMSE usually assign higher weights for large errors than smaller errors.

$$MAE = \frac{\sum_{i=1}^n |\hat{y}_i - y_i|}{n} \quad (4.2)$$

The MAE deals with errors uniformly based on their sizes. It is the average of the absolute values of the estimation errors.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (4.3)$$

Thus apart from squared coefficient correlation R^2 which measures correlation between the actual and predicted values that must have a reasonable goodness of fit well above 50%, the model with the lowest MAE and RMSE is considered a better method. Again another important measure of model performance used is the Akaike's Information Criterion (AIC). This accuracy test statistic is designed to choose from many competing models the optimal model based on the maximum likelihood criterion of the model

parameter. McCluskey & Borst (2011: 304) report that the estimates of β_i are based on the least squares and the maximum likelihood estimates of the model parameters are identical. This permit the expression of AIC in terms of statistics available from the OLS regression as:

$$AIC = n + n \ln(2\pi) + n \log\left(\frac{RSS}{n}\right) + 2K \quad (4.4)$$

where K is the estimated parameters in the model that include the intercept and σ^2 (McCluskey & Borst, 2011: 304) while the RSS denotes the sample residual sum of squares. The model with the minimum AIC is adjudged the best in this study.

Again to measure models quality in line with consistency and uniformity, the benchmark test acceptable to the International Association of Assessing Officers (IAAO, 2013) was used. These are coefficient of dispersion (COD) and price related differential (PRD). The COD measures assessment uniformity and variability of how much the value ratios vary from the median ratio.

$$COD = \frac{\sum |Z_i - Z_m|}{nZ} \times 100 \quad (4.5)$$

PRD on the other hand is used to measure consistency of valuation ratios between low valued properties and high valued properties. In essence this technique measures the vertical equity of appraisals which should be less than 1.0 (progressivity), because measures above 1.0 suggest regressivity (IAAO, 2013: 29).

$$PRD = \frac{\frac{\sum Z_i}{n}}{\frac{\sum PV_i}{\sum AV_i}} \quad (4.5)$$

where, Z_i is used to depicts PV_i/AV_i ; and PV_i is the predicted value of the i th property; AV_i is the assessed value; Z_i is the ratio between the predicted value and assessed value of the i th property and Z_m is the median of Z_i .

5.0 Empirical analysis and discussions of results

This section contains result of the hedonic regression models and the various ANNs training algorithms. The model building was done in categories with a view to selecting the optimal model within the same category before a general comparative analysis on the best two performed models is carried out to select the best model based on the set criteria.

5.1 Establishing a baseline model for the Cape Town property market

The study first undertakes the test for neglected nonlinearities in the linear model to establish the case for using the nonlinear techniques.

5.2 Test for neglected nonlinearities

Peterson & Flanagan (2009: 158) report that those supporting the use of ANNs based their argument on the neglected nonlinearities observed in the linear models. Consequently, a RESET test that applied the predictions of the ANNs as applied by Peterson & Flanagan (2009); and McCluskey *et al.* (2012) was used to justify the application of nonlinear models. The predictions of ANNs are included as an additional regressor in the OLS. Therefore if the OLS specification is:

$$y = X\beta + \mu \quad (5.1)$$

and the predictions of the ANNs are contained in the vector m , then a test of neglected nonlinearities is equivalent to a t -test on $H_0: \varphi = 0$ in the regression:

$$\mu = X\beta + \varphi m + v \quad (5.2)$$

The null of no neglected nonlinearities is easily rejected at any conventional level of significance which reveals the need for using the nonlinear techniques in the data. Table 5.1 provides the summary.

Table 5.1 OLS test for neglected nonlinearities

Variable	Coefficients	Std. coefficient	VIF	Std. error	T	P Value
Constant	184128.64	.000	0.000	108996.235	1.689	.091
ANNs_PRED	.169	.302	5.434	.020	8.279	.000
Beds1	-268912.33	-.016	1.029	268324.101	-1.002	.316
Beds2	59294.49	.012	1.158	86323.676	.687	.492
Beds4	-15482.94	-.005	1.350	55025.949	-.281	.778
Beds5	84891.11	.018	1.294	84908.868	1.000	.317
Beds6	-41474.57	-.005	1.193	140961.806	-.294	.769
Beds7	-199621.72	-.011	1.069	300670.729	-.664	.507
Beds8	-718658.19	-.023	1.048	489621.563	-1.468	.142
Beds9	-563217.54	-.010	1.014	900162.925	-.626	.532
Beds10	-1611303.8	-.020	1.023	1278560.17	-1.260	.208
Poor quality	281408.02	.008	1.898	779474.055	.361	.718
Fair quality	282547.11	.018	1.268	270141.889	1.046	.296
Good quality	120315.50	.042	2.572	71753.480	1.677	.094
Very good quality	692168.80	.049	1.145	236679.916	2.924	.003
Excellent quality	224460.42	.018	1.391	226330.294	.992	.321
Poor condition	671187.54	.022	1.901	659403.960	1.018	.309
Fair condition	10129.23	.001	1.265	213404.801	.047	.962
Good condition	-21277.10	-.007	2.507	71271.145	-.299	.765
Excel. condition	51451.23	.008	1.524	124717.101	.413	.680
Storey_2	1740.85	.001	1.500	54616.097	.032	.975
Storey_3	180262.75	.021	1.369	158106.703	1.140	.254
Sub-economic style	-5557.41	.000	1.020	737259.479	-.008	.994
Unconventional style	-23656.01	-.003	1.262	157940.060	-.150	.881
Georgian victor style	208584.53	.016	1.035	202106.906	1.032	.302
Cape Dutch style	-304658.87	-.014	1.019	354544.091	-.859	.390
Maisonette style	108745.01	.008	1.042	213405.683	.510	.610
Mediterranean style	-40112.35	-.001	1.034	575263.505	-.070	.944
Partly obstructed view	-50059.37	-.007	1.091	109684.505	-.456	.648

Below average view	-32586.80	-.002	1.019	293559.725	-.111	.912
Above average view	153332.36	.051	1.283	53134.430	2.886	.004
Panoramic view	343931.27	.085	1.510	77445.606	4.441	.000
Excellent view	-11128.77	-.001	1.127	185853.749	-.060	.952
Smkt48	-256405.29	-.020	1.158	213843.225	-1.199	.231
Smkt50	-173579.83	-.031	1.492	106282.625	-1.633	.103
Smkt52	-245923.90	-.065	1.821	79270.687	-3.102	.002
Smkt53	-88541.59	-.021	1.565	84031.206	-1.054	.292
Smkt55	519394.25	.097	2.317	127637.310	4.069	.000
Smkt56	158524.51	.031	1.529	100060.322	1.584	.113
Smkt64	-291337.73	-.012	1.034	371839.580	-.784	.433
Smkt65	-43415.82	-.001	1.008	897615.767	-.048	.961
Smkt66	-150383.35	-.019	1.183	133362.619	-1.128	.260
Smkt67	151260.73	.033	1.574	90234.646	1.676	.094
Smkt68	243594.95	.043	1.422	105282.361	2.314	.021
Smkt69	-110396.79	-.008	1.156	221757.391	-.498	.619
Smkt70	-275218.52	-.028	1.303	174352.198	-1.579	.115
Smkt73	-196250.27	-.004	1.011	734343.680	-.267	.789
SA1	22488.97	.006	1.440	69159.491	.325	.745
SA2	-75546.55	-.021	1.466	67391.449	-1.121	.262
SA3	-60514.74	-.017	1.456	68269.379	-.886	.375
SA5	1074.55	.000	1.424	69578.385	.015	.988
Size	2.778	.000	2.155	413.752	.007	.995
Pool	-599.422	-.008	1.316	1389.581	-.431	.666
Dependent variable = Residual		AIC = 100046.796				

5.3 The baseline regression model

The linear, semi-log and log-log models are tested based on their R^2 , adjusted R^2 , AIC and F -statistics to show levels of their statistical acceptability and confidence. Apart from the RESET test undertaken above, the study of McCluskey (2016: 130) observes that variables transformation into their log form makes it possible to explore the presence of nonlinear relationship in the data.

Table 5.2 Goodness-of-fit measurements for linear, semi-log and log-log models

Model	Regression		
	Linear	Semi-log	Log-log
R^2	0.589	0.694	0.687
Adjusted R^2	0.583	0.689	0.682
F -statistic	89.441	141.527	137.045
AIC	103057.028	1461.717	1533.562

The R^2 statistic shows the explanation of variation in property prices across the linear, semi-log and log-log models. Accordingly the parameter estimates reveal that 58.9%, 69.4% and 68.7% of the variation in property prices are explained by the three models respectively. There was a reduction in the level of explanation to 58.3%, 68.9% and 68.2% when adjustment was made to the general population as revealed in the adjusted R^2 statistic. The adjusted R^2 revealed acceptability of the results since the maximum value is 100%. Again, the three regression models exhibit high F -statistics at $P < 0.01$; linear regression (89.441); semi-log (141.527); and log-log (137.045). In all the semi-log outperform the other regression models (linear and log-log). However, the goodness of fit criterion used in Table 5.2 was not sufficient to establish

a baseline regression model for the city of Cape Town because the three models have different structure. The COD as used by Borst (2007: 160) can be used for direct comparison of the three models (Table 5.3). Additional measures of model performance including Median Ratio, Mean Ratio, RMSE, MAE and Price Related Differentials are also used. The mean and median ratios are part of the quality assurance benchmark tests recommended for industry used. The standard stipulates that a ratio of between 0.90 and 1.10 meets the required standards.

Table 5.3 Performance comparison of linear, semi-log and log-log models

Test	Regression		
	Linear	Semi-log	Log-log
<i>Median ratio</i>	1.055	1.027	1.032
<i>Mean ratio</i>	1.087	1.044	1.046
<i>PRD</i>	1.08	1.10	1.10
<i>COD</i>	26.7	22.1	22.6
<i>MAE</i>	1234015	1091961	1127104
<i>RMSE</i>	1997911	1951184	1995487

The results in Table 5.3, reveal the semi-log model (1951184) to perform better in comparison to linear and log-log models (1997911 and 1995487) in terms of RMSE. According to Limsombunchai, *et al.* (2004: 196), a model with the lowest RMSE is considered the best in terms of prediction accuracy. The MAE value also reveals the semi-log model (1091961) to predict prices that are closer to the assessed values than the linear and log-log models (1234015 and 1127104). The results show marginal performance of the log-log to linear models in terms of predicting property prices closer to the assessed values. Furthermore, the semi-log model has the best performance in terms of the mean ratio, median ratio and COD while the only slim advantage of the linear model was that it had a slightly better PRD relating to the vertical equity of prices. Therefore on the strength of these results the baseline model for the Cape Town property market is the semi-log model.

The detailed results reveal the regression coefficients, t-statistics and indicators of level of significance of the three regression based models. This is presented in Table 5.4, Table 5.5 and Table 5.6. The test of multicollinearity using the variance inflation factor (VIF) indicates that in all assessments (linear, semi-log and log-log), there is no inflation of any of the regression coefficient.

Table 5.4 Linear regression model coefficients

Variable	Coefficients	Std. coefficient	VIF	Std. error	T	P Value
Constant	1925246.24			158212.73	12.169	.000
Beds1	-95531.07	-.003	1.029	427538.96	-.223	.823
Beds2	15630.32	.001	1.158	137541.94	.114	.910
Beds4	37641.80	.006	1.350	87676.96	.429	.668
Beds5	291784.11	.028	1.293	135278.58	2.157	.031
Beds6	629564.89	.035	1.182	223563.63	2.816	.005
Beds7	637132.87	.016	1.066	478398.83	1.332	.183
Beds8	5007485.25	.075	1.017	768486.58	6.516	.000
Beds9	171405.83	.001	1.013	1434258.1	.120	.905
Beds10	3867852.90	.022	1.015	2029838.2	1.905	.057
Poor quality	-202524.20	-.003	1.897	1241655.6	-.163	.870
Fair quality	-61100.14	-.002	1.267	430244.67	-.142	.887
Good quality	442444.77	.071	2.537	113563.08	3.896	.000
Very good quality	2618454.27	.084	1.123	373413.86	7.012	.000

Excellent quality	2756016.95	.102	1.282	346187.56	7.961	.000
Poor condition	363167.64	.005	1.900	1050457.9	.346	.730
Fair condition	-52365.85	-.002	1.264	339991.77	-.154	.878
Good condition	247138.27	.039	2.505	113526.24	2.177	.030
Excel. condition	-209527.37	-.015	1.519	198402.30	-1.056	.291
Storey_2	613462.18	.098	1.377	83371.75	7.358	.000
Storey_3	2392759.12	.126	1.264	242111.40	9.883	.000
Sub-economic style	280819.94	.003	1.019	1174492.0	.239	.811
Unconventional style	2103625.07	.107	1.143	239491.88	8.784	.000
Georgian victor style	572529.82	.021	1.034	321948.83	1.778	.075
Cape Dutch style	-286716.32	-.006	1.018	564814.73	-.508	.612
Maisonette style	92098.37	.003	1.042	340034.39	.271	.787
Mediterranean style	-38174.25	.000	1.034	916498.59	-.042	.967
Partially obstructed view	176691.38	.012	1.090	174756.93	1.011	.312
Below average view	-139569.28	-.003	1.019	467739.72	-.298	.765
Above average view	485613.16	.074	1.268	84179.59	5.769	.000
Panoramic view	1327260.85	.150	1.382	118050.99	11.243	.000
Excellent view	1994698.18	.081	1.075	289304.63	6.895	.000
Smkt48	-2515927.1	-.090	1.089	330435.18	-7.614	.000
Smkt50	-2094033.0	-.172	1.281	156943.99	-13.343	.000
Smkt52	-1316265.2	-.160	1.603	118522.76	-11.106	.000
Smkt53	-1136953.0	-.121	1.424	127714.31	-8.902	.000
Smkt55	3852948.92	.328	1.460	161459.66	23.863	.000
Smkt56	1391301.39	.123	1.439	154642.04	8.997	.000
Smkt64	-2176342.9	-.042	1.023	589279.11	-3.693	.000
Smkt65	-1892688.7	-.015	1.006	1429077.4	-1.324	.185
Smkt66	-830327.69	-.048	1.166	210958.29	-3.936	.000
Smkt67	-874903.05	-.087	1.463	138579.95	-6.313	.000
Smkt68	-445484.83	-.036	1.373	164824.71	-2.703	.007
Smkt69	-2032349.3	-.070	1.095	343996.14	-5.908	.000
Smkt70	-2293589.6	-.107	1.206	267271.65	-8.581	.000
Smkt73	-1698941.2	-.017	1.008	1168266.1	-1.454	.146
SA1	-35065.4	-.004	1.439	110172.13	-.318	.750
SA2	-187413.1	-.024	1.463	107292.86	-1.747	.081
SA3	-42573.42	-.005	1.455	108746.12	-.391	.695
SA5	132591.65	.016	1.423	110819.77	1.196	.232
Size	9156.69	.232	1.789	600.68	15.244	.000
Pool	7025.48	.041	1.300	2200.61	3.193	.001
Dependent variable: Assessed values						

The regression coefficients show the contribution of specific attributes to property price in the linear model. The result shows that property size is positively related to property price at R9,156.69. The base number of bedroom is a three bedroom property. It therefore follows that properties with two, four, five, six, seven, eight and ten bedrooms are positively related to property prices while properties with one and nine bedrooms are negatively related to price in this study. The quality grade factor and condition of properties are essential. As expected, properties having poor and fair qualities are negatively related to price while good, very good and excellent properties are positively related to prices. The condition of properties provides a result that is somewhat counterintuitive in excellent condition because of the negative values. While the negative value for fair condition is expected, the same cannot be said of excellent property condition which should ordinarily be additive towards property price, albeit at an appreciating rate. The number of storeys is very important, and a two and three storey properties are associated with higher prices.

The building style reveals that sub-economic, unconventional, Georgian victor, and maisonette are positively related property price, while Cape Dutch and Mediterranean building style are negatively related to price. Property view is also very important, and coefficients show that properties that are partially obstructed, above average, panoramic and excellent views are positively related to price while properties having view that is below average is negatively related to price. The location of a property is very important in determination of its price. Consequently, the submarket dummy variable is used to depict areas where the property is located. Properties that are located in SMKT55 and SMKT56 are positively related to property prices while all properties located in the remaining submarkets are negatively related to price. The base time of sale is in the fourth (SA4) semi-annual notation. Properties that are sold in the fifth semi-annual time of sale (SA5) are positively related to price with R132,591.65, while properties sold during the first (SA1), second (SA2) and third (SA3) semi-annual time of sale are negatively related to price. Also properties that have a swimming pool are positively related to price with R7,025.48 than those properties without a swimming pool.

Table 5.5 Semi-log regression model coefficients

Variable	Coefficients	Std. coefficient	VIF	Std. error	T	P Value
Constant	14.722			.024	622.852	.000
Beds1	-.086	-.013	1.029	.064	-1.352	.176
Beds2	-.034	-.018	1.158	.021	-1.675	.094
Beds4	.038	.033	1.350	.013	2.909	.004
Beds5	.071	.039	1.293	.020	3.522	.000
Beds6	.092	.029	1.182	.033	2.754	.006
Beds7	.118	.017	1.066	.071	1.649	.099
Beds8	.524	.045	1.017	.115	4.562	.000
Beds9	.085	.004	1.013	.214	.399	.690
Beds10	.201	.007	1.015	.303	.663	.507
Poor quality	-.080	-.006	1.897	.185	-.433	.665
Fair quality	-.002	.000	1.267	.064	-.036	.972
Good quality	.089	.082	2.537	.017	5.227	.000
Very good quality	.266	.050	1.123	.056	4.773	.000
Excellent quality	.350	.075	1.282	.052	6.768	.000
Poor condition	.058	.005	1.900	.157	.371	.711
Fair condition	-.019	-.004	1.264	.051	-.379	.705
Good condition	.048	.044	2.505	.017	2.843	.004
Excel. condition	-.066	-.027	1.519	.030	-2.240	.025
Storey_2	.148	.137	1.377	.012	11.904	.000
Storey_3	.342	.104	1.264	.036	9.458	.000
Sub-economic style	.070	.004	1.019	.175	.396	.692
Unconventional style	.169	.050	1.143	.036	4.722	.000
Georgian victor style	.117	.024	1.034	.048	2.439	.015
Cape Dutch style	-.077	-.009	1.018	.084	-.910	.363
Maisonette style	-.006	-.001	1.042	.051	-.125	.900
Mediterranean style	.233	.017	1.034	.137	1.700	.089
Partially obstructed view	.100	.039	1.090	.026	3.820	.000
Below average view	-.019	-.003	1.019	.070	-.265	.791
Above average view	.115	.101	1.268	.013	9.184	.000
Panoramic view	.242	.158	1.382	.018	13.720	.000
Excellent view	.287	.068	1.075	.043	6.638	.000
Smkt48	-.914	-.190	1.089	.049	-18.520	.000

Smkt50	-.570	-.270	1.281	.023	-24.299	.000
Smkt52	-.339	-.237	1.603	.018	-19.123	.000
Smkt53	-.248	-.152	1.424	.019	-13.000	.000
Smkt55	.502	.247	1.460	.024	20.800	.000
Smkt56	.202	.103	1.439	.023	8.737	.000
Smkt64	-.810	-.091	1.023	.088	-9.200	.000
Smkt65	-.609	-.028	1.006	.213	-2.851	.004
Smkt66	-.205	-.069	1.166	.032	-6.492	.000
Smkt67	-.220	-.126	1.463	.021	-10.635	.000
Smkt68	-.120	-.056	1.373	.025	-4.874	.000
Smkt69	-.733	-.147	1.095	.051	-14.273	.000
Smkt70	-.829	-.224	1.206	.040	-20.763	.000
Smkt73	-1.024	-.058	1.008	.175	-5.868	.000
SA1	-.014	-.010	1.439	.016	-.845	.398
SA2	-.035	-.026	1.463	.016	-2.157	.031
SA3	-.021	-.016	1.455	.016	-1.310	.190
SA5	.003	.002	1.423	.017	.206	.837
Size	.002	.271	1.789	.000	20.674	.000
Pool	.001	.050	1.300	.000	4.451	.000

Dependent variable: Ln Assessed values

The regression coefficients of the semi-log and log-log models reveal that a one and two bedrooms properties are negatively related to property prices while properties with higher number of bedrooms are positively related to price. Again as expected, properties with poor and fair quality grade factor are negatively related to price while higher quality properties are positively related to price. Properties with poor and good conditions are positively related to price while those of fair and excellent conditions are negatively related to price. Again this is counterintuitive as already noted in the negative sign in the coefficients of excellent properties in the linear models. Also properties with more than two storeys are positively related to price. Property size and swimming pool all have positive correlation on price in the semi-log model. Similar variables have positive influence on property price in the log-log model. The results show the three regression models have dissimilar significant variables. The linear model has the least number of significant variables (28); the semi-log model has a slightly higher number of significant variables (34) and the log-log model has the highest number of significant variables (35). Again, quite a number of variables have the appropriate signs in the nonlinear regression models than in linear regression.

Table 5.6 Log-log regression model coefficients

Variable	Coefficients	Std. coefficient	VIF	Std. error	T	P Value
Constant	13.492			.083	162.123	.000
Beds1	-.076	-.008	1.034	.093	-.811	.417
Beds2	-.032	-.011	1.171	.030	-1.063	.288
Beds4	.056	.034	1.363	.019	2.917	.004
Beds5	.117	.045	1.291	.029	3.989	.000
Beds6	.177	.039	1.169	.048	3.652	.000
Beds7	.216	.021	1.062	.104	2.076	.038
Beds8	.754	.045	1.017	.167	4.504	.000
Beds9	.282	.009	1.010	.312	.903	.367
Beds10	.678	.015	1.010	.441	1.536	.125
Poor quality	-.120	-.006	1.898	.271	-.445	.656
Fair quality	.008	.001	1.266	.094	.090	.928

Good quality	.133	.085	2.540	.025	5.356	.000
Very good quality	.380	.049	1.124	.081	4.672	.000
Excellent quality	.527	.078	1.282	.075	6.990	.000
Poor condition	.087	.005	1.900	.229	.379	.705
Fair condition	-.038	-.006	1.264	.074	-.510	.610
Good condition	.066	.042	2.509	.025	2.646	.008
Excel. condition	-.100	-.028	1.519	.043	-2.311	.021
Storey_2	.216	.138	1.411	.018	11.739	.000
Storey_3	.500	.106	1.276	.053	9.432	.000
Sub-economic style	.128	.005	1.020	.256	.502	.616
Unconventional style	.242	.049	1.144	.052	4.644	.000
Georgian victor style	.162	.023	1.035	.070	2.311	.021
Cape Dutch style	-.080	-.006	1.016	.123	-.648	.517
Maisonette style	.053	.007	1.051	.074	.712	.476
Mediterranean style	.273	.014	1.035	.200	1.364	.173
Partially obstructed view	.130	.035	1.089	.038	3.425	.001
Below average view	-.036	-.004	1.018	.102	-.355	.722
Above average view	.171	.104	1.268	.018	9.329	.000
Panoramic view	.357	.162	1.385	.026	13.856	.000
Excellent view	.437	.071	1.078	.063	6.924	.000
Smkt48	-1.300	-.187	1.094	.072	-18.019	.000
Smkt50	-.813	-.267	1.280	.034	-23.775	.000
Smkt52	-.482	-.235	1.612	.026	-18.624	.000
Smkt53	-.344	-.146	1.424	.028	-12.374	.000
Smkt55	.722	.246	1.460	.035	20.510	.000
Smkt56	.357	.126	1.392	.033	10.768	.000
Smkt64	-1.169	-.091	1.024	.128	-9.097	.000
Smkt65	-.874	-.028	1.006	.311	-2.805	.005
Smkt66	-.297	-.069	1.165	.046	-6.453	.000
Smkt67	-.305	-.121	1.460	.030	-10.114	.000
Smkt68	-.153	-.050	1.371	.036	-4.266	.000
Smkt69	-.961	-.133	1.120	.076	-12.685	.000
Smkt70	-1.117	-.209	1.241	.059	-18.907	.000
Smkt73	-1.452	-.057	1.009	.255	-5.700	.000
SA1	-.026	-.013	1.439	.024	-1.072	.284
SA2	-.056	-.029	1.462	.023	-2.409	.016
SA3	-.033	-.017	1.456	.024	-1.413	.158
SA5	-.001	.000	1.423	.024	-.031	.976
Size	.304	.259	1.899	.016	18.946	.000
Pool	.014	.044	1.318	.004	3.855	.000
Dependent variable: Ln Assessed values						

In general the results of the log transformed regression are better than the linear regression, thereby improving the predictive accuracy. The study of McCluskey (2016: 135) observed that it is imperative to have quality control measures such as the log transformation to confirm the results of a predictive model before such estimates are used for taxation. This analysis shows that the semi-log regression models can be used to achieve the objective of tax assessors in arriving at a tax amount for properties in Cape Town. As noted earlier, due to the limitations of the hedonic regression models various improvements were made to enhance its predictive accuracy. The next section of this analysis is concern with the use of the ANNs designed to overcome the rigidity of the parametric restrictions of the hedonic regression models.

5.4 The influence of ANNs training algorithms in mass appraisal

The structure of the ANNs is different from the regression based techniques. There is no *a priori* pre-specification of the relationship between the dependent and independent variables. The variables used for ANNs training and testing are identical but having differing composition as previously noted. Four ANNs training algorithms including BP, SCG, PBCG and LM were used in this assessment. The results are presented in Table 5.7. The performance of models reveals a very good square correlation between the actual assessed values and the estimated values for the all data, training and testing datasets. For the BP training algorithm, the R^2 reveals that the ANNs architecture explains 67.9% of prices for all data, 66.2% for training data and 77.5% for testing data. There is improvement in the performance of the SCG, PBCG and LM relative to the BP in explaining the variation in property prices. Specifically, the SCG algorithm explains 75.9%, 75.6% and 76.5% of variation in property prices for all data, training and testing data. Also the R^2 statistic measures for PBCG revealed 75%, 75.1% and 74.9% variation in property prices for all data, training and testing data. The LM revealed variance in property price as shown in their R^2 , 75.8%, 79.7% and 67.2% for all data, training and testing datasets, respectively.

The MAE accuracy test statistic revealed the LM algorithm to predict property prices closer to the assessed values than all other training algorithms for all data, training and testing data (239298, 243832 and 228724). The BP training algorithm performed worse in the all data and testing data (1057708 and 1013821) but have slight improvement in the training data (1288025) as revealed in the MAE accuracy test. BP algorithm slightly outperforms the SCG in the training data. The RMSE accuracy test shows the BP algorithm to perform poorly in all data and test data in comparison to the other training algorithms. The BP algorithm also marginally outperforms the SCG in the training dataset. The LM reveals a lower RMSE in the training and testing dataset, thus showing that in terms of model prediction and accuracy it is better than other training models.

Table 5.7 Goodness-of-fit measurements and prediction comparison of BP, LM, PBCG and SCG algorithms

Performance measures		R^2	Mean ratio	Median ratio	PRD	COD %	MAE	RMSE
SCG	All data	0.7590	1,0263	0,9941	1,0275	11.77	256469,9	397494,6
	Train data	0.7558	1,0251	0,9940	1,0277	11.82	1762084	2889345
	Test data	0.7653	1,0291	0,9965	1,0272	11.62	253126,3	407919,9
PBCG	All data	0.7504	1,0276	0,9978	1,0257	11.75	255564,8	527264,9
	Train data	0.7509	1,0279	0,9979	1,0262	11.85	257894,6	571605,3
	Test data	0.7495	1,0269	0,9978	1,0245	11.50	250131,6	405444,6
LM	All data	0.7583	1,0129	0,9957	1,0229	10.77	239297,9	513824,2
	Train data	0.7973	1,0139	0,9936	1,0239	10.98	243832,4	559611,0
	Test data	0.6716	1,0106	1,0003	1,0207	10.26	228723,7	386524,8
BP	All data	0.6789	1,0461	1,0278	1,0829	25.14	1057708	1773281
	Train data	0.6618	0,8262	0,7951	1,0180	28.76	1288025	2042443
	Test data	0.7749	1,0859	1,0413	1,0606	24.16	1013821	1410072

The results of the different ANN training algorithms when viewed in the light of the IAAO approved guidelines for mass appraisal reveal that apart from the BP training algorithm, all other models demonstrate consistency between lower valued and higher valued properties. The PRD for all other models and the training data of BP reveals that it lie between 1.01–1.03, depicting that neither the lower and higher valued properties are favoured. However the result shows that the BP algorithm performs better in one scenario – the training data than the PRD of training data of other models. Again, analysis shows that the LM algorithm performs better in the training and testing datasets in terms of their COD statistic but the BP algorithm

performs poorly in terms of uniformity and horizontal or random dispersion (COD). The analysis reveals a high similarity of results among the different algorithms, therefore to arrive at a definite conclusion on which training algorithm perform optimal, a reliability ranking order as used in McCluskey, *et al.* (2013: 258) was employed. The analysis was carried out on the R^2 , PRD, COD, MAE and RMSE for all data, training and testing datasets Table 5.8.

Table 5.8 Performance of algorithms and reliability ranking order

Training algorithms	R^2	PRD	COD	MAE	RMSE	Overall rank
All data						
SCG	1	3	2	3	1	2.0
PBCG	3	2	1	2	3	2.2
LM	2	1	2	1	2	1.6
BP	4	4	4	4	4	4.0
Training data						
SCG	2	4	2	4	4	3.2
PBCG	3	3	3	2	2	2.6
LM	1	2	1	1	1	1.2
BP	4	1	4	3	3	3.0
Testing data						
SCG	2	3	3	3	3	2.8
PBCG	3	2	2	2	2	2.2
LM	4	1	1	1	1	1.6
BP	1	4	4	4	4	3.4

The results of reliability ranking order in Table 5.8 reveal the LM training algorithm to outperform all other models in this study. The LM algorithm has the properties of gradient descent and Newton speed which enables it to train the ANNs faster and better than the most widely used BP algorithm. Although the SCG and PBCG algorithms perform well in this study they nonetheless could not surpass the LM training algorithm. The next section concerns the comparison between LMANNs which perform best among the training algorithm and the baseline regression model.

5.5 Comparison of the performance of LMANNs and semi-log models

Previous studies are not wholly conclusive in terms of model superiority. The study of McCluskey *et al.* (2012: 285) observes that the right question on model superiority is not rightly asked. The correct question should be “which modelling approach meets the rigorous standards of transparency, stability of output, predictive ability and defensibility for the mass appraisal industry?” This question is worthy of used in this study to select the optimal model for the Cape Town property market. The analyses reveal the baseline model as the semi-log, and the optimal artificial neural networks training algorithm as the Levenberg-Marquardt. The tests used for comparison are the model performance and reliability ranking order, model explainability ranking order (McCluskey *et al.* 2013: 259) and prediction accuracy of a model within 10 and 20% of the actual assessed values. The model that performs best in all measures is selected as the Cape Town mass appraisal model.

The first test is to assess the level of prediction accuracy which falls within 10 and 20% of the assessed values as used by Thibodeau (2003: 17); McCluskey, *et al.* (2013: 255). The results are presented in Table 5.9. The overall results reveal that semi-log model fail the minimum standard of 50% prediction accuracy within 10% of the assessed values but the LMANNs marginally achieve the benchmark of 50%. These models have a good predictive performance. There is a slight improvement in the semi-log model when the benchmark was increase to prediction within 20% of the assessed values. In all, the LMANNs performed

best in this test. The model has a good prediction accuracy of 53.1% property appraised within 10% and 57.1% predicted values falling within 20% of the assessed values.

Table 5.9 Model prediction within 10 and 20% of actual assessed values

Model	Percent within 10%	Percent within 20%
<i>Semi-log</i>	43.4%	50.5%
<i>LMANNs</i>	51.1%	57.1%

The second test employed is the performance and reliability ranking. The R^2 , median and mean ratios, COD, PRD, MAE and RMSE accuracy measures were used. The accuracy measures were selected because they are found in all platforms used. The results are summarised in table 5.10.

Table 5.10 General model performance and reliability ranking order

Accuracy measures	Semi-log Rank	LMANNs Rank
R^2	2	1
<i>Median ratio</i>	2	1
<i>Mean ratio</i>	2	1
<i>PRD</i>	2	1
<i>COD</i>	2	1
<i>MAE</i>	2	1
<i>RMSE</i>	2	1
<i>Overall rank</i>	2.00	1.00

The results presented in Table 5.10 are tied to the Cape Town property data. The comparative analysis reveals the Levenberg-Marquardt trained artificial neural networks (LMANNs) to outperform the semi-log model. The LMANNs has both gradient descent search method and Newton speed which enhances its performance. The overall result favours the LMANNs as the best in terms of predictive accuracy but it should be noted that within the mass appraisal field, equally significant is the ability to support the assessed/estimated values before a tribunal (McCluskey *et al.*, 2012: 284). Although such challenges against the model rarely occur, the detail estimates predicted should reasonably be comprehensible to the appraiser such that in the occasion of any challenge this should explicitly be explain.

Table 5.11 provides the results of explicit ability of a model to explain details which is very important in mass appraisal setting. The hedonic regression based model has the history of being used within the mass appraisal environment despite its shortcoming because of the simple and consistent manner it provides statistical evidence which can help in objective judgement about superiority of the predictive method in terms of R^2 , adjusted R^2 , F -statistics, t -value and levels of significance of variables. Again McCluskey *et al.* (2013: 258) noted that the traditional hedonic regression models are preferred because of the single set of parameter estimates which is consistently applied. These qualities of the traditional regression models give it an edge over the other model used in this comparative analysis, hence, the semi-log model occupy a unique position one as shown in Table 5.11. In terms of explicit nature of location,

Table 5.11 Model explicit explainability ranking order

Measurement Criteria	Semi-log Rank	LMANNs Rank
<i>Simple</i>	1	2
<i>Consistent</i>	1	2
<i>Transparency</i>	1	2
<i>Locational</i>	1	2
<i>Applicability</i>	1	2
<i>Mean ranking</i>	1.00	2.00

the traditional models (semi-log) is not inherently locational (McCluskey *et al.*, 2013: 259). The models (linear, semi-log and log-log) do not capture explicit location (x, y coordinates) but locational dummies can be used. The locational dummies could be neighbourhood, political or administrative units or more precisely known submarkets. The ANNs have ability to recognised patterns; hence, locational dummies are equally used. Additionally, McCluskey *et al.* (2013: 259) opine that absolute location can also be used for ANNs but there is no such evidence in this study. The most important measure is the applicability of models in the mass appraisal environment. It is not enough for a model to accurately predict values but to what extent can the model be used within the mass appraisal environment. Table 5.9 and Table 5.10 reveal the LMANNs to be the best performed model in terms of prediction accuracy but results in Table 5.11 negates this finding. This is because of the non-transparent nature of the estimates predicted by the model. The black box nature of the ANNs makes it difficult to implement within the mass appraisal field. The hedonic regression models, particularly, the semi-log form can be implemented for mass appraisal because of the explicit manner the parameter estimates are provided.

5.0 Concluding remarks

The importance of modelling property prices that is devoid of the many parametric restrictions of the hedonic regression has resulted in a plethora of academic research over the last 40 years. The BP training algorithm of the ANNs is one of such models that have been applied in modelling price with varying degrees of success, although, few studies took a contrary position that negates such claim. This study represents a first attempt to evaluate the feasibility and effectiveness of other ANNs training algorithms in modelling property prices and compare results with the hedonic regression models. Using the industry and standard benchmark accuracy statistical tests, the results reveal that PBCG, LM and SCG algorithms outperform the traditional BP algorithm in this study with the overall performance been credited to the LM training algorithm. The LMANNs has the ability of combining both gradient descent and Newton speed, thus enhancing its speed in training the ANNs. In general, the ANNs do not require much effort in data pre-processing. They do not require creation of binary or transformation to linearise variables and over training of fitting is less of a current issue with the models. They do however suffer from non-transparency in interpretation of coefficients. Again the trial and error procedure of ANNs can create slightly inconsistent and unreliable results each time the technique is used.

The hedonic regression models have long been used to provide a well-established approach in the mass appraisal domain. In contrast to the ANNs, the regression models create coefficients for each variable providing a transparent platform for explanation and defence should the appraisal exercise be call to question. The ability of the regression based models to replicate findings irrespective of the software utilised is an added advantage of the model. The results of this study reveal that while the LMANNs perform best in terms of prediction accuracy within the 10 and 20% of the assessed values and performance reliability ranking order, the model was rejected on the basis of non-transparency and applicability within the mass appraisal environment. The results favour the semi-log model which though in the context of predictive accuracy was below the LMANNs but preferred as the best in the contexts of simplicity, consistency, locational, transparency and applicability in the mass appraisal domain

For future research, it might be of interest to improve the degree of explainability of ANNs through a research that promote the opening of the black box (to see what is happening inside) so that parameter estimates can be viewed explicitly. Until this is done these models would continue to be regarded as techniques that supports and complements econometric analysis and not as substitutes. Again data from other jurisdictions within the South African property market would be useful to determine the robustness of the findings. It will also be of great interest to utilise property data from other regions within the sub-Saharan Africa to ascertain the robustness of the models. Finally, to add voice to what the study of McCluskey *et al.* (2012: 286) opined having considered the computing abilities of the regression techniques and the ANNs that a hybrid of the duo should be formed such that the results from ANNs could be used to smoothing the output generated by the hedonic regression models. This is because despite the high performance of other training algorithms in this study, the results did not make any difference with the back propagation training algorithm in terms of transparency.

ACKNOWLEDGEMENT

Opinions expressed in this study are those of the authors. Financial assistance received from the National Research Foundation (NRF) and the IRE/BS Foundation for African Real Estate Research is hereby acknowledged. We also thank Justine Kahonde a staff of city valuation office Cape for the property data and other useful information provided.

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